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— TWO —

A Brief Visit to the Systems Zoo

> The … goal of all theory is to make the … basic elements as simple and as few as possible without having to surrender the adequate representation of … experience.
>
> —Albert Einstein,[1] physicist

One good way to learn something new is through specific examples rather than abstractions and generalities, so here are several common, simple but important examples of systems that are useful to understand in their own right and that will illustrate many general principles of complex systems.

This collection has some of the same strengths and weaknesses as a zoo.[2] It gives you an idea of the large variety of systems that exist in the world, but it is far from a complete representation of that variety. It groups the animals by family—monkeys here, bears there (single-stock systems here, two-stock systems there)—so you can observe the characteristic behaviors of monkeys, as opposed to bears. But, like a zoo, this collection is too neat. To make the animals visible and understandable, it separates them from each other and from their normal concealing environment. Just as zoo animals more naturally occur mixed together in ecosystems, so the systems animals described here normally connect and interact with each other and with others not illustrated here—all making up the buzzing, hooting, chirping, changing complexity in which we live.

Ecosystems come later. For the moment, let's look at one system animal at a time.

One-Stock Systems

A Stock with Two Competing Balancing Loops—a Thermostat

You already have seen the "homing in" behavior of the goal-seeking balancing feedback loop—the coffee cup cooling. What happens if there are two such loops, trying to drag a single stock toward two different goals?

One example of such a system is the thermostat mechanism that regulates the heating of your room (or cooling, if it is connected to an air conditioner instead of a furnace). Like all models, the representation of a thermostat in Figure 15 is a simplification of a real home heating system.

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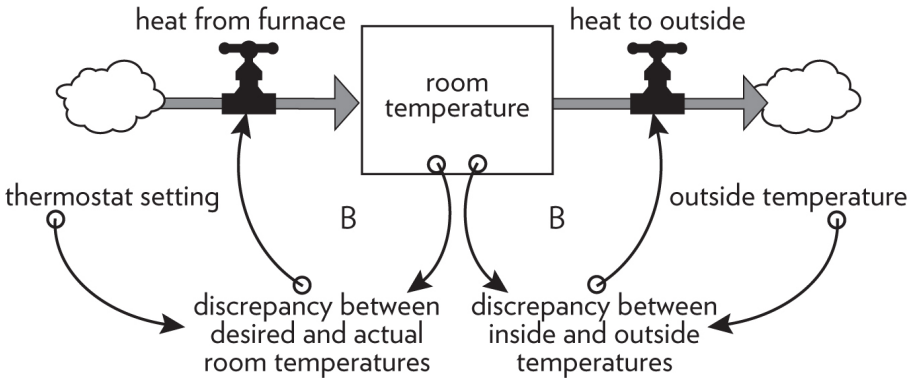


Figure 15. Room temperature regulated by a thermostat and furnace.

Whenever the room temperature falls below the thermostat setting, the thermostat detects a discrepancy and sends a signal that turns on the heat flow from the furnace, warming the room. When the room temperature rises again, the thermostat turns off the heat flow. This straightforward, stock-maintaining, balancing feedback loop is shown on the left side of Figure 15. If there were nothing else in the system, and if you start with a cold room with the thermostat set at 18°C (65°F), it would behave as shown in Figure 16. The furnace comes on, and the room warms up. When the room temperature reaches the thermostat setting, the furnace goes off, and the room stays right at the target temperature.

However, this is not the only loop in the system. Heat also leaks to the outside. The outflow of heat is governed by the second balancing feedback loop, shown on the right side of Figure 15. It is always trying to make the room temperature equal to the outside, just like a coffee cup cooling. If this were the only loop in the system (if there were no furnace), Figure 17 shows what would happen, starting with a warm room on a cold day.

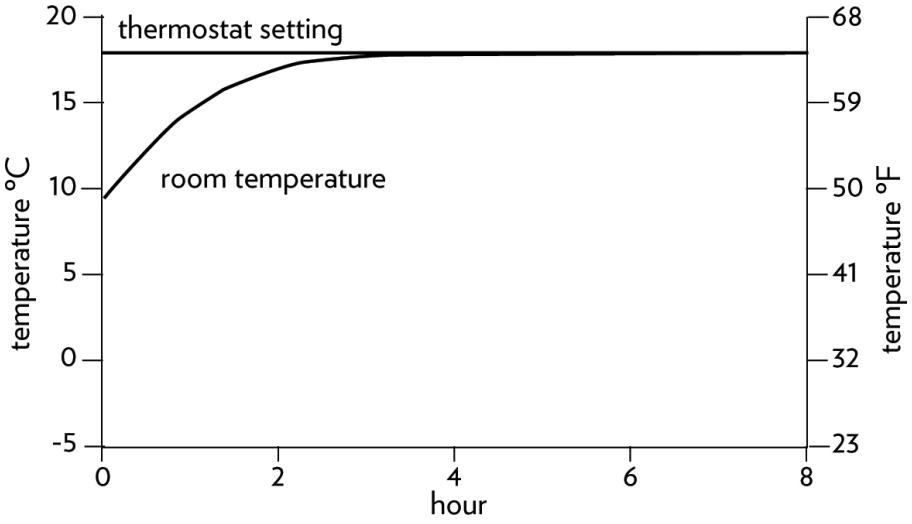


Figure 16. A cold room warms quickly to the thermostat setting.

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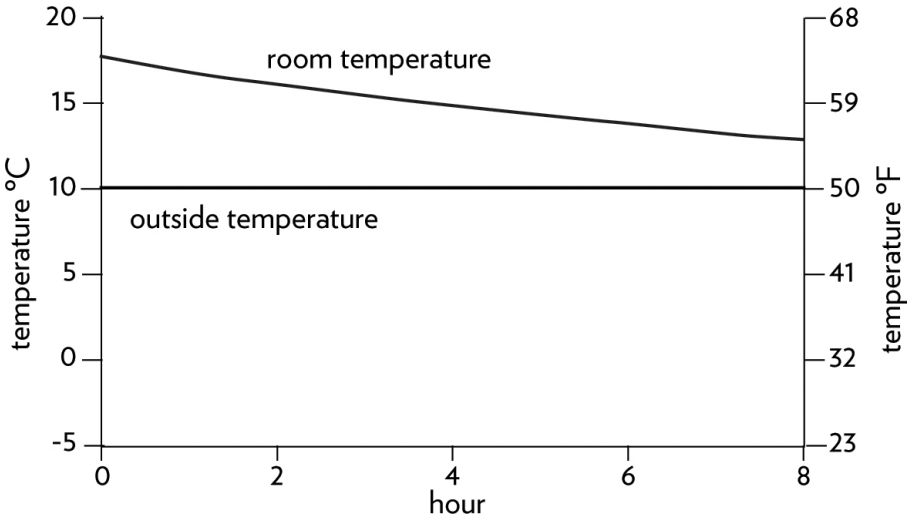


Figure 17. A warm room cools very slowly to the outside temperature of 10°C.

The assumption is that room insulation is not perfect, and so some heat leaks out of the warm room to the cool outdoors. The better the insulation, the slower the drop in temperature would be.

Now, what happens when these two loops operate at the same time? Assuming that there is sufficient insulation and a properly sized furnace, the heating loop dominates the cooling loop. You end up with a warm room (see Figure 18), even starting with a cold room on a cold day.

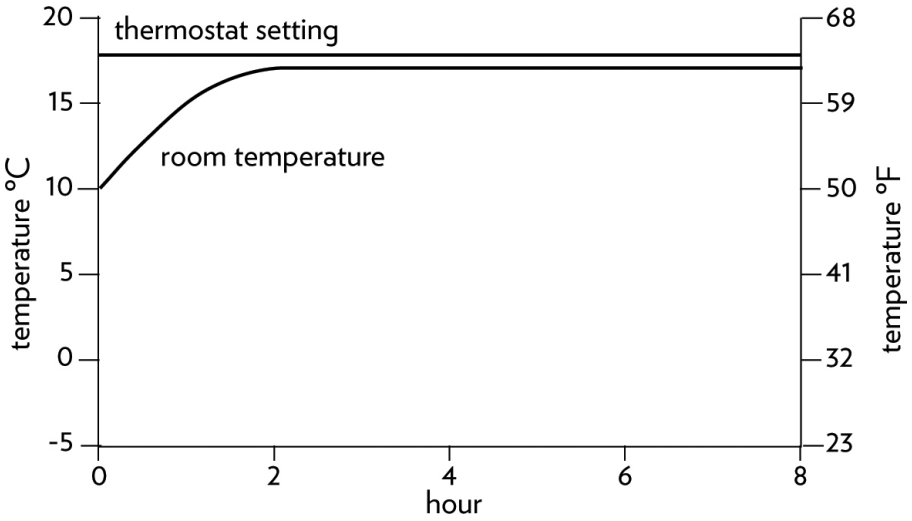


Figure 18. The furnace warms a cool room, even as heat continues to leak from the room.

As the room heats up, the heat flowing out of it increases, because there's a larger gap between inside and outside temperatures. But the furnace keeps putting in more heat than the amount that leaks out, so the room warms nearly to the target temperature. At that point, the furnace cycles off and on as it compensates for the heat constantly flowing out of the room.

The thermostat is set at 18°C (65°F) in this simulation, but the room temperature levels off slightly below 18°C (65°F). That's because of the leak to the outside, which is draining away some heat even as the furnace is getting the signal to put it back. This is a characteristic and sometimes surprising behavior of a system with competing balancing loops. It's like trying to keep a bucket full when there's a hole in the bottom. To make things worse, water leaking out of the hole is governed by a feedback loop; the more water in the bucket, the more the water pressure at the hole increases, so the flow out increases! In this case, we are trying to keep the room warmer than the outside and the warmer the room is, the faster it loses heat to the outside. It takes time for the furnace to correct for the increased heat loss—and in that minute still more heat leaks out. In a well-insulated house, the leak will be slower and so the house more comfortable than a poorly insulated one, even a poorly insulated house with a big furnace.

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With home heating systems, people have learned to set the thermostat slightly higher than the actual temperature they are aiming at. Exactly how much higher can be a tricky question, because the outflow rate is higher on cold days than on warm days. But for thermostats this control problem isn't serious. You can muddle your way to a thermostat setting you can live with.

For other systems with this same structure of competing balancing loops, the fact that the stock goes on changing while you're trying to control it can create real problems. For example, suppose you're trying to maintain a store inventory at a certain level. You can't instantly order new stock to make up an immediately apparent shortfall. If you don't account for the goods that will be sold while you are waiting for the order to come in, your inventory will never be quite high enough. You can be fooled in the same way if you're trying to maintain a cash balance at a certain level, or the level of water in a reservoir, or the concentration of a chemical in a continuously flowing reaction system.

There's an important general principle here, and also one specific to the thermostat structure. First the general one: The information delivered by a feedback loop can only affect future behavior; it can't deliver the information, and so can't have an impact fast enough to correct behavior that drove the current feedback. A person in the system who makes a decision based on the feedback can't change the behavior of the system that drove the current feedback; the decisions he or she makes will affect only future behavior.

The information delivered by a feedback loop—even nonphysical feedback—can only affect future behavior; it can't deliver a signal fast enough to correct behavior that drove the current feedback. Even nonphysical information takes time to feedback into the system.

Why is that important? Because it means there will always be delays in responding. It says that a flow can't react instantly to a flow. It can react only to a change in a stock, and only after a slight delay to register the incoming information. In the bathtub, it takes a split second of time to assess the depth of the water and decide to adjust the flows. Many economic models make a mistake in this matter by assuming that consumption or production can respond immediately, say, to a change in price. That's one of the reasons why real economies tend not to behave exactly like many economic models.

The specific principle you can deduce from this simple system is that you must remember in thermostat-like systems to take into account whatever draining or filling processes are going on. If you don't, you won't achieve the target level of your stock. If you want your room temperature to be at 18°C (65°F), you have to set the thermostat a little above the desired temperature. If you want to pay off your credit card (or the national debt), you have to raise your repayment rate high enough to cover the charges you incur while you're paying (including interest). If you're gearing up your work force to a higher level, you have to hire fast enough to correct for those who quit while you are hiring. In other words, your mental model of the system needs to include all the important flows, or you will be surprised by the system's behavior.

A stock-maintaining balancing feedback loop must have its goal set appropriately to compensate for draining or inflowing processes that affect that stock. Otherwise, the feedback process will fall short of or exceed the target for the stock.

Before we leave the thermostat, we should see how it behaves with a varying outside temperature. Figure 19 shows a twenty-four-hour period of normal operation of a well-functioning thermostat system, with the outside temperature dipping well below freezing. The inflow of heat from the furnace nicely tracks the outflow of heat to the outside. The temperature in the room varies hardly at all once the room has warmed up.

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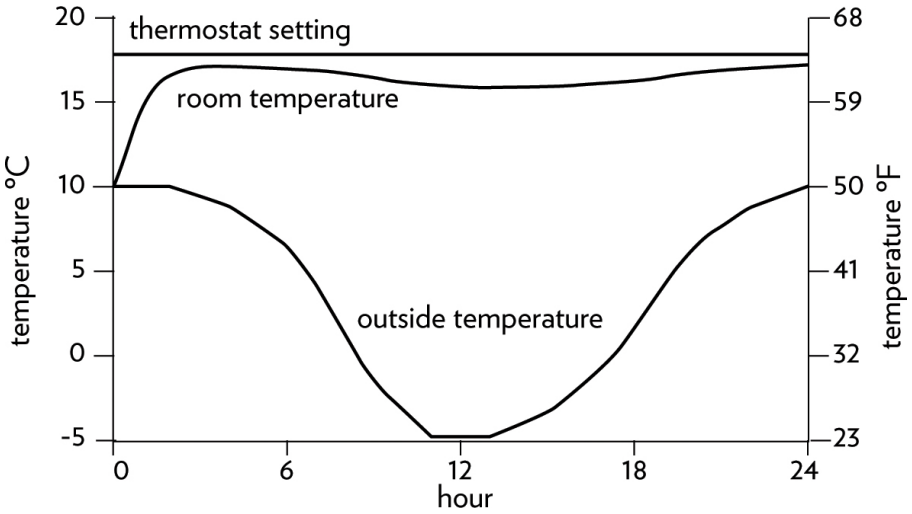


Figure 19. The furnace warms a cool room, even as heat leaks from the room and outside temperatures drop below freezing.

Every balancing feedback loop has its breakdown point, where other loops pull the stock away from its goal more strongly than it can pull back. That can happen in this simulated thermostat system, if I weaken the power of the heating loop (a smaller furnace that cannot put out as much heat), or if I strengthen the power of the cooling loop (colder outside temperature, less insulation, or larger leaks). Figure 20 shows what happens with the same outside temperatures as in Figure 19, but with faster heat loss from the room. At very cold temperatures, the furnace just can't keep up with the heat drain. The loop that is trying to bring the room temperature down to the outside temperature dominates the system for a while. The room gets pretty uncomfortable!

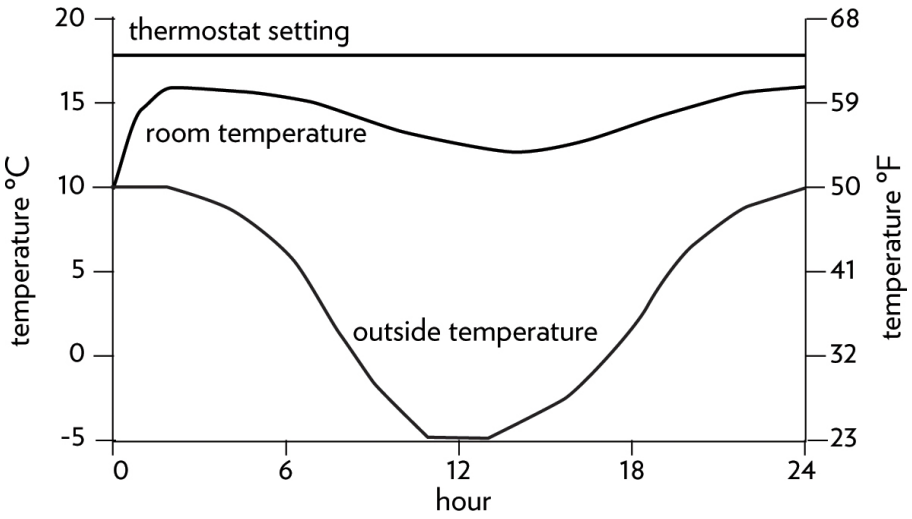


Figure 20. On a cold day, the furnace can't keep the room warm in this leaky house!

See if you can follow, as time unfolds, how the variables in Figure 20 relate to one another. At first, both the room and the outside temperatures are cool. The inflow of heat from the furnace exceeds the leak to the outside, and the room warms up. For an hour or two, the outside is mild enough that the furnace replaces most of the heat that's lost to the outside, and the room temperature stays near the desired temperature.

But as the outside temperature falls and the heat leak increases, the furnace cannot replace the heat fast enough. Because the furnace is generating less heat than is leaking out, the room temperature falls. Finally, the outside temperature rises again, the heat leak slows, and the furnace, still operating at full tilt, finally can pull ahead and start to warm the room again.

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Just as in the rules for the bathtub, whenever the furnace is putting in more heat than is leaking out, the room temperature rises. Whenever the inflow rate falls behind the outflow rate, the temperature falls. If you study the system changes on this graph for a while and relate them to the feedback-loop diagram of this system, you'll get a good sense of how the structural interconnections of this system—its two feedback loops and how they shift in strength relative to each other—lead to the unfolding of the system's behavior over time.

A Stock with One Reinforcing Loop and One Balancing Loop—Population and Industrial Economy

What happens when a reinforcing and a balancing loop are both pulling on the same stock? This is one of the most common and important system structures. Among other things, it describes every living population and every economy.

A population has a reinforcing loop causing it to grow through its birth rate, and a balancing loop causing it to die off through its death rate.

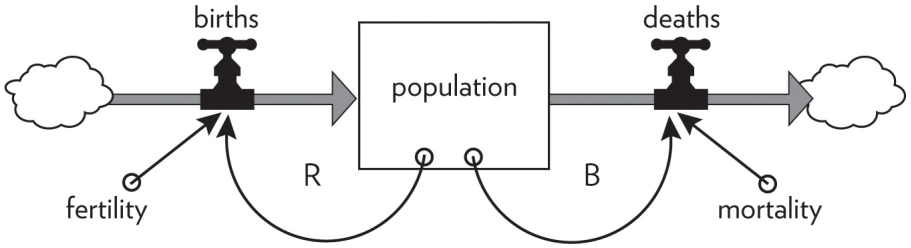


Figure 21. Population governed by a reinforcing loop of births and a balancing loop of deaths.

As long as fertility and mortality are constant (which in real systems they rarely are), this system has a simple behavior. It grows exponentially or dies off, depending on whether its reinforcing feedback loop determining births is stronger than its balancing feedback loop determining deaths.

For example, the 2007 world population of 6.6 billion people had a fertility rate of roughly 21 births a year for every 1,000 people in the population. Its mortality rate was 9 deaths a year out of every 1,000 people. Fertility was higher than mortality, so the reinforcing loop dominated the system. If those fertility and mortality rates continue unchanged, a child born now will see the world population more than double by the time he or she reaches the age of 60, as shown in Figure 22.

If, because of a terrible disease, the mortality rate were higher, say at 30 deaths per 1,000, while the fertility rate remained at 21, then the death loop would dominate the system. More people would die each year than would be born, and the population would gradually decrease (Figure 23).

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